

PCT/PCM 8: The Kernel That Learned

4. Interpret the Figure

The learned kernel produces a better result than the fixed hand-designed sharpening kernel, with the figure showing about 33.8 dB versus 33.0 dB PSNR. Its 3×3 weights have a much stronger center and stronger negative surrounding weights (approximately center 10.37, adjacent values -4.33 , and corner values about 1.86), allowing it to learn a more suitable sharpening pattern from the data.

5. Parameter Calculation

For 1 input channel, 8 output channels, 3×3 kernels and one bias per output channel:

Parameters per output channel = $(3 \times 3 \times 1) + 1 = 10$.

Total = $10 \times 8 = 80$ parameters.

For 64 input channels and 64 output channels:

Parameters per output channel = $(3 \times 3 \times 64) + 1 = 577$.

Total = $577 \times 64 = 36,928$ parameters.

The number of weight connections increases by $64 \times 64 / (1 \times 8) = 512 \times$. Including biases, the total parameter count increases from 80 to 36,928.

6. Prediction

If the clean target is a blurred copy of the input, the learned 3×3 kernel should behave like a smoothing/blur kernel. Its nine weights would therefore be mostly positive, with the center generally having a relatively large weight and the surrounding weights distributing the averaging effect. The weights would approximately sum to 1 for brightness-preserving averaging, although the exact values depend on the training data.